

Observe that we could have written our statement using two different variables x and y , as $\exists x (P(x) \wedge Q(x)) \vee \exists y R(y)$, because the scopes of the two quantifiers do not overlap.

1.4.9 Negating Quantified Expressions

- Consider the negation of the statement "Every student in your class has taken a course in calculus."
- This statement is a universal quantification, namely, $\forall x P(x)$, where $P(x)$ is the statement " x has taken a course in calculus" and the domain consists of the students in your class. The negation of this statement is "It is not the case that every student in your class has taken a course in calculus." This is equivalent to "There is a student in your class who has not taken a course in calculus." And this is simply the existential quantification of the negation of the original propositional function, namely $\exists x \neg P(x)$.

This example illustrates the following logical equivalence: $\boxed{\neg \forall x P(x) \equiv \exists x \neg P(x)}$

To show that $\neg \forall x P(x)$ and $\exists x \neg P(x)$ are logically equivalent no matter what the propositional function $P(x)$ is and what the domain is, first note that $\neg \forall x P(x)$ is true if and only if $\forall x P(x)$ is False. Next note that $\forall x P(x)$ is False if and only if there is an element x in the domain for which $P(x)$ is False. This holds if and only if there is an element x in the domain for which $\neg P(x)$ is True. Finally, note that there is an element x in the domain for which $\neg P(x)$ is true if and only if $\exists x \neg P(x)$ is true. It follows that $\neg \forall x P(x)$ and $\exists x \neg P(x)$ are logically equivalent since we can conclude that $\neg \forall x P(x)$ is true if and only if $\exists x \neg P(x)$ is true.

Suppose we wish to negate an existential quantification. For instance, consider the proposition "There is a student in this class who has taken a course in calculus." This is the existential quantification $\exists x Q(x)$ where $Q(x)$ is the statement " x has taken a course in calculus." The negation of this